Danesh Zeynali

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I am a computer systems researcher working on Internet transport—congestion control, low-latency networking, and Internet measurement. My approach is experimental: I characterize how protocols actually behave in the Internet, and I design, build, and evaluate networked systems grounded in those characterizations. Six years as a networking engineer before my doctorate, and ongoing collaborations with industry, draw me to fundamental problems whose answers matter for deployment. I aim to develop my academic career by designing and teaching systems courses and building an independent research agenda.

Professional and Academic Experience

- **Max Planck Institute for Informatics** — **Saarbrücken, Germany**
  Graduate Research Assistant — *Oct 2020–Present*
- **Nokia Bell Labs** — **Antwerp, Belgium**
  Ph.D. Research Intern — *Dec 2025–Mar 2026*
- **MTN Irancell** — **Tehran, Iran**
  Packet Core Optimization Engineer — *Jan 2019–Sep 2020*
- **Rightel** — **Tehran, Iran**
  Packet Core Planning Engineer — *Sep 2016–Dec 2018*
- **PARMAN** — **Tehran, Iran**
  Network Management System Developer — *Sep 2014–Aug 2016*
- **Iran Telecommunication Research Center** — **Tehran, Iran**
  Research Intern — *Jun 2009–Sep 2009*

Education

- **Max Planck Institute for Informatics, Saarland University** — **Saarbrücken, Germany**
  Ph.D. Candidate, Computer Science — *2020–2026 (expected)*
  - Advisor: Prof. Anja Feldmann
  - Co-Advisor: Prof. Balakrishnan Chandrasekaran
- **Sharif University of Technology** — **Tehran, Iran**
  M.Sc., Information Technology – Computer Networks — *2012–2015*
- **Institute for Advanced Studies in Basic Sciences (IASBS)** — **Zanjan, Iran**
  B.Sc., Information Technology — *2007–2012*

Publications

D. Zeynali, B. Chandrasekaran, and A. Feldmann, "L4S Congestion Control Under Dynamic Workloads," in *ACM Internet Measurement Conference (IMC) Student Workshop*, 2026

D. Zeynali, C. Chang, K. De Schepper, B. Chandrasekaran, and A. Feldmann, "Small Packets, Big Difference: L4S Congestion Control at Low Bandwidth," in *Applied Networking Research Workshop (ANRW)*, 2026

D. Zeynali, D. Eminov, T. Albakour, B. Chandrasekaran, and A. Feldmann, "The State of ECN Adoption in the Internet," in *Network Traffic Measurement and Analysis Conference (TMA)*, 2026

S. Fathalli, E. N. Weyulu, **D. Zeynali**, B. Chandrasekaran, and A. Feldmann, "Network Assisted Congestion Feedback," *IEEE Transactions on Network and Service Management (TNSM)*, 2025

D. Zeynali, E. N. Weyulu, S. Fathalli, B. Chandrasekaran, and A. Feldmann, "Promises and Potential of BBRv3," in *Passive and Active Measurement (PAM)*, 2024. 🏆 *Best Paper Award*

D. Zeynali, E. N. Weyulu, S. Fathalli, B. Chandrasekaran, and A. Feldmann, "BBRv3 in the Public Internet: A Boon or a Bane?," in *Applied Networking Research Workshop (ANRW)*, 2024

T. Shreedhar, **D. Zeynali**, O. Gasser, N. Mohan, and J. Ott, "A Longitudinal View at the Adoption of Multipath TCP," 2022. arXiv preprint

D. Zeynali, F. Osali, and A. M. A. Hemmatyar, "Geographic Routing With Energy Constraint in Wireless Multimedia Sensor Networks," in *27th Iranian Conference on Electrical Engineering (ICEE)*, 2019

Supervision

- **Evaluating Congestion Control with Different QUIC Implementations and Streams**
  Master's Thesis – Maham Shakir, Saarland University *2026*
- **L4S Mousetrap: Extending L4S to Understand Flow Size**
  Master's Thesis – Jeremy G. Diamond, Vrije Universiteit Amsterdam *2025*
- **ECN Adoption in the Internet**
  Bachelor's Thesis – Denis Eminov, Vrije Universiteit Amsterdam *2025*
  Resulted in a publication at TMA 2026.
- **Comprehensive Evaluation of BBRv2 and BBRv3 in the NS-3 Simulator**
  Master's Thesis – Felix Brinkmann, Saarland University *2024*
- **Evaluating the Performance Impact of ECN on BBRv3 Using the NS-3 Simulator**
  Bachelor's Thesis – Julian Dietz, Saarland University *2024*

Teaching Experience

- **Hot Topics in Data Networks Seminar**
  Organizer and Supervisor, Saarland University *2020–2025*
- **Data Networks**
  Tutor, Saarland University *2020–2022*
- **Wireless Networks**
  Tutor, Sharif University of Technology *2014*
- **Database Systems**
  Tutor, Institute for Advanced Studies in Basic Sciences (IASBS) *2009*

Talks and Presentations

- **Network Traffic Measurement and Analysis Conference (TMA)**
  The State of ECN Adoption in the Internet *2026*
- **Applied Networking Research Workshop (ANRW)**
  Small Packets, Big Difference: L4S Congestion Control at Low Bandwidth *2026*
- **Applied Networking Research Workshop (ANRW)**
  BBRv3 in the Public Internet: A Boon or a Bane? *2024*

Awards, Grants, and Academic Service

- **RACI Grant Recipient**, RIPE 93, 2026
- **Best Paper Award**, Passive and Active Measurement Conference (PAM), 2024
- **RIPE Fellowship**, RIPE 79
- **Shadow TPC Member**, ACM Internet Measurement Conference (IMC), 2022 and 2025

Technical Expertise

- **Programming:** C++, Python (data analysis and visualization), P4 (programmable data planes), Shell scripting, Java, SQL
- **Networking Tools:** NS-3 (congestion control design and evaluation), Linux networking tools (tcpdump, Ipsumdump, tc, iproute2), Wireshark/Tshark, Zeek
- **Networking Knowledge:** CCNP-level routing and switching, TCP/IP, VLANs, Software-Defined Networking (SDN), and network troubleshooting
- **Telecommunication Technologies:** 2G, 3G, and 4G/LTE packet-core technologies

References

Prof. Anja Feldmann
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Saarbrücken, Germany
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Prof. Balakrishnan Chandrasekaran
Vrije Universiteit Amsterdam
Amsterdam, Netherlands
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